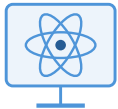


Multi-fidelity problem

self-driving laboratory

candidate pool $\mathcal{X}_{\text{cand}}$



Low fidelity

simulation
cheap, abundant · $\rho \ll 1$



High fidelity

experiment
costly, scarce · cost 1

Controlled comparison

fix budget · schedule · capacity

→ **vary only the surrogate**

Gaussian-process
(GP) family

MFGP · NARGP
DKL · sparse GP

Transfer-learning
(TL) surrogates $\times 11$

$g^{(L)}$

$g^{(H)}$

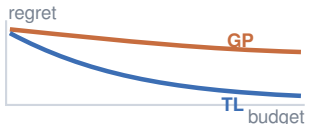
+

$\hat{y}^{(H)}$

residual transfer + Bayesian linear-regression (BLR) head

Transfer-learning wins

on molecular & materials tasks



converges fast, up to $16\times$ lower regret

compute per optimization loop



TL **up to $45\times$ fewer FLOPs**

GP TL

closed-loop optimization: propose next query, evaluate, update